

Proof-of-Continuity: A Temporal Model for Authority Propagation in Distributed Systems and AI Agents

Nicola Gallo

8 July 2026

Abstract

Proof-of-Possession authorization models derive authority from the possession of artifacts such as tokens, credentials, or capabilities. This paper argues that possession is insufficient for discrete execution chains, whether they span multiple services or occur as separated steps within the same machine, because it does not guarantee preservation of the causal relationship between the origin of a request and the authority exercised at later steps. We introduce Proof-of-Continuity, a minimal authority-propagation discipline for the Provenance Identity Continuity (PIC) model, in which each execution step must be causally linked to the previous step and may only propagate a non-expansive subset of the authority received from the origin. Under this model, the confused deputy condition cannot be satisfied as valid model behavior: any privilege exercised at a later step must already be present in the origin authority context. This is directly relevant to distributed systems and AI agents, where executors invoke tools and downstream services while holding multiple authority sources, so that the same authority/causality mismatch recurs across service boundaries. Under Proof-of-Continuity these sources may be carried together but are never merged into a combined authority, since each step is authorized only against the authority context of the lineage that caused it.

This paper concerns authorization propagation rather than authentication: identity and authentication mechanisms such as OIDC, verifiable credentials, wallets, and workload identity remain complementary mechanisms for establishing the origin, while Proof-of-Continuity addresses how authority propagates after that origin exists.

1 Model

We formalize the PIC (Provenance Identity Continuity) Model as follows.

Let P be a finite set of principals, O a set of operations, and R a set of resources. Define a privilege as $(o, r) \in O \times R$.

Each principal p has an associated privilege set:

$$Priv(p) \subseteq O \times R.$$

A principal is used here as an abstract permissioned entity: it may be a human identity established through an identity provider, wallet, OIDC/OIDC4VC-style flow [13, 14], or decentralized identifier [15]; a workload or machine identity such as those used in WIMSE/SPIFFE-style environments [11, 12]; a role; a service account; or another authenticated permissioned entity. What matters for this model is that a privilege set $Priv(p)$ is associated with it and that it can express an intent within those privileges. When a permissioned entity p expresses such an intent, it selects a subset of $Priv(p)$; that subset becomes the origin authority context $C_0 \subseteq Priv(p)$ for a new execution lineage.

For example, suppose Bob holds the privileges (read, Book) and (write, Book). When Bob expresses an intent he selects a subset — say only (write, Book) — and that subset becomes the origin authority context C_0 of a new lineage. Two executions may both involve Bob and the same selected privilege, yet need not be the same occurrence: each carries authority propagated along its own lineage. One may be a valid continuation of an intent Bob expressed within a lineage; another may be an unrelated execution exercising the same apparent privilege outside the lineage that granted it. The identity and privilege coordinates may be identical; the lineage coordinate distinguishes the two.

A request is a message *req* passed between executors — the entities that perform execution steps — and may contain a payload.

Execution is modeled as a causal chain of steps:

$$\pi = \langle (s_0, C_0), (s_1, C_1), \dots, (s_n, C_n) \rangle$$

where s_0 is the origin step and $C_i \subseteq O \times R$ is the authority context carried at step s_i . We use s_i for the formal execution step at hop i ; operationally, a hop is the execution of one such step by an executor that receives, processes, and may propagate a request. Each hop may be performed by a service, workload, function, tool, or agent; the model requires continuity of authority, not propagation of identity.

Definition 1 (PIC Model). *A system enforces Provenance Identity Continuity if, for every execution chain π , each transition preserves causal continuity and monotonic authority restriction:*

$$C_{i+1} \subseteq C_i$$

for all i , and every privilege exercised at hop n is causally linked to the origin via a verifiable chain.

The single-hop causal link asserted at each transition is a *Proof of Relationship (PoR)*: evidence that a step is a valid continuation of its immediate predecessor within the same lineage. The PIC Model thus combines two conditions — causal linkage, witnessed hop-by-hop by *PoR*, and monotonic restriction $C_{i+1} \subseteq C_i$ — both introduced informally here and defined formally in Section 5, where continuity is built as the transitive composition of these relationships. The theorem below states the authority consequence of the second condition.

Theorem 1 (PIC Safety). *If the PIC Model holds, no execution step can exercise a privilege not present at the origin.*

Proof. By construction, $C_{i+1} \subseteq C_i$ for all i , so $C_n \subseteq C_0$. Thus, any $(o, r) \in C_n$ must also be in C_0 . Therefore, no privilege can be gained beyond the origin authority context.

2 Threat Model and Scope

We consider execution chains in which an origin request causes one or more later execution steps. Executors may be buggy, confused, or adversarially influenced through their inputs or their execution context. An executor may hold multiple authority sources — ambient privileges, delegated tokens, bearer credentials, or capabilities — and may attempt to use one unrelated to the request being processed.

The adversary may influence requests, payloads, and parameters passed across the chain, and may cause an executor to select the wrong authority source. The adversary cannot forge the relationship evidence required by *PoR*, nor make a valid continuity checker accept a transition that violates monotonicity. As capability and token models assume their credentials unforgeable, we assume the relationship evidence and the authority-transition mechanism unforgeable; compromise of the validator, the cryptographic root of trust, or the initial authority bootstrap is out of scope.

3 Confused Deputy

Definition 2 (Confused Deputy). *A confused deputy occurs when there exist principals U (user) and D (deputy) and a privilege $(o, r) \in O \times R$ such that:*

1. $(o, r) \notin \text{Priv}(U)$,
2. $(o, r) \in \text{Priv}(D)$,
3. U sends a request req to D ,
4. as a consequence of req , D executes (o, r) .

This definition does not depend on implementation details, only on the mismatch of authority and causality.

Classical example. A FORTRAN compiler FORT, installed in the privileged directory SYSX, holds ambient authority to write statistics to (SYSX)STAT. A user invokes the compiler and provides (SYSX)BILL as the debugging output filename. The compiler opens the target for output using its own authority over SYSX, thereby overwriting (SYSX)BILL. The user never possessed this privilege, but the deputy exercised it as a consequence of the user's request [1].

4 Proof-of-Possession (PoP)

A token t represents a set of privileges:

$$\text{Priv}(t) \subseteq O \times R.$$

PoP Semantics. Possession implies usability: if a principal holds t , it may exercise all $(o, r) \in \text{Priv}(t)$.

PoP systems constrain authority by artifact possession, but do not, by themselves, require that the exercised authority be a causal continuation of the authority context of the request.

4.1 Sender-Constrained and Holder-Bound Tokens

Sender-constrained and holder-bound tokens — for example DPoP [9] and Token Binding [10] — prove *who* may present or use an artifact by binding it to a key or transport channel. They do not, by themselves, prove that the authority exercised through that artifact is a continuation of the current

execution lineage. Restricting who may present a token constrains transport, not causality: a legitimate holder can still, at some step, exercise a possessed authority that lies outside the lineage of the request it is processing. The binding answers holder authenticity; the lineage question — whether this use continues the authority context that caused it — remains open.

4.2 Authority Mixing

Assume:

$$(\text{write}, r) \notin \text{Priv}(U) \quad (\text{H1})$$

$$(\text{write}, r) \in \text{Priv}(D) \quad (\text{H2})$$

Here U is the user and D is a deputy.

In a possession-based model, a deputy may hold multiple authority sources: its own ambient privileges, delegated tokens, bearer credentials, or other artifacts. The model authorizes an operation by checking whether some possessed authority source grants the requested privilege.

Definition 3 (Selection function). *Let an executor's held authority be a family $\{A_j\}$ of sources — ambient privileges, delegated tokens, and bound or sender-constrained artifacts. A selection function σ chooses, for a requested (o, r) , some source A_j with $(o, r) \in A_j$, independently of the request lineage ℓ — the causal lineage of the request being processed, formalized in Section 5.*

Theorem 2 (PoP admits authority mixing). *In a possession-based model, if an executor may select authority from any artifact or ambient privilege it holds independently of the request authority context, then the confused deputy condition is admissible in the model.*

Proof. Let a request originate from an authority context that does not contain (o, r) . Let the executor hold another authority context, artifact, or ambient privilege that does contain (o, r) .

In a PoP model, authorization depends on the possession of an artifact or privilege granting (o, r) , not on whether that authority is a continuation of the request authority context. Therefore the model permits an execution in which the request causally triggers (o, r) while (o, r) is absent from the request authority context. This is the confused deputy condition of Definition 2.

This applies to any model in which authorization is derived from possession independently of the causal lineage, including JWTs, bearer credentials, and capabilities treated as held artifacts when they may be applied outside

the execution that requested them. Holder binding and sender-constraint restrict which principal may present A_j , not whether σ 's choice lies in ℓ ; they therefore do not remove this admissibility. The gap is not possession itself but the absence of a temporal binding between the authority used and the execution that caused its use.

These are execution-level conditions: authority belonging to one lineage is drawn into another, producing the confused deputy. Proof-of-Continuity removes the admissibility at the level of the model — in a valid PIC execution the confused-deputy condition cannot arise (Section 7), because authority is defined only as a continuation of the causing lineage.

5 Proof-of-Continuity (PoC / PIC)

Execution is a causal chain:

$$s_0 \rightarrow s_1 \rightarrow \cdots \rightarrow s_n.$$

Let $C_i \subseteq O \times R$ denote the authority context carried by execution step s_i .

Definition 4 (Proof of Relationship). *PoR(s_i, s_{i+1}) holds iff s_{i+1} is a valid causal continuation of s_i within the same execution lineage. PoR is a relational, single-hop property: it binds one step to its immediate predecessor. The reach back to the origin is not carried by any single PoR but obtained by composing consecutive relationships, as the definition of PoC below makes precise.*

Definition 5 (Valid transition). *A transition from s_i to s_{i+1} is valid iff it is both causally related and non-expansive:*

$$\text{Valid}(s_i, C_i, s_{i+1}, C_{i+1}) \iff \text{PoR}(s_i, s_{i+1}) \wedge C_{i+1} \subseteq C_i.$$

The first condition proves causal relationship. The second condition proves monotonic authority restriction. Thus each hop transfers only:

$$C_{i+1} \subseteq C_i. \tag{C1}$$

Definition 6 (Proof of Continuity). *For a chain $\pi = \langle s_0, \dots, s_n \rangle$, PoC(π) holds iff every adjacent transition is valid:*

$$\text{PoC}(\pi) \iff \bigwedge_{i=0}^{n-1} (\text{PoR}(s_i, s_{i+1}) \wedge C_{i+1} \subseteq C_i).$$

Thus continuity is the transitive composition of relationships: *PoR* proves a single link, *PoC* proves the whole lineage. Relationship is local; continuity is global.

Monotone propagation along a causal chain. Two things propagate together. First, authority attenuates monotonically: write $C_{i+1} \preceq C_i$ for authority *attenuation* — every authority at a hop is a restriction of one held at the previous hop, never an expansion. When the context is a set of atomic privileges this specializes to $C_{i+1} \subseteq C_i$, the form used in our examples. Although the core model represents authority contexts as sets of atomic privileges, we use $\preceq$ when referring to the general attenuation order; in the set-valued case used throughout the formal examples, this order is $\subseteq$. Second, the hops are bound into a single causal lineage by *PoR*: continuity is a property not of a hop in isolation but of the *chain*. Formally the lineage is a logical causal chain; in an implementation it need not be carried whole — it may be represented by cryptographic commitments, references, hashes, signatures, or summarized proofs that allow each hop to prove its place in the lineage without transporting the entire history.

Embedding execution in the authority model. Proof-of-Continuity is defined precisely for this setting: the successor step need not be known, selected, instantiated, or provisioned when the predecessor completes its step. Execution is a sequence of causal steps in time, not of positions fixed by topology — the upstream executor acts at some time t , while the downstream executor may come into existence only at a later time $t' > t$. Pre-binding authority to a concrete holder, key, or channel of an unknown successor is therefore not always possible.

What must hold across the gap is not the identity of the successor, nor a policy attached to it, but that its authority continues the context that caused it. A credential, workload-identity assertion, service-account credential, or bound token may authenticate the holder or bind an artifact to a key, workload, or channel. Unless the authorization decision also reads lineage-sensitive evidence, however, the same holder attributes may appear across distinct executions: they establish who acts, not which causal chain produced this request. Re-deriving authority at $i + 1$ from a fresh assertion about the successor, rather than from its relationship to the causing step, reintroduces the individuation gap of Section 5.3 — the structural condition behind the confused deputy (Definition 2). Continuity avoids this because it is not self-assertable: defined only relative to the predecessor, it can be

demonstrated as a continuation but not asserted at $i + 1$ in isolation.

A predecessor may instead emit a continuation later consumed by some successor:

$$(s_i, C_i) \xrightarrow{\text{emit}} \text{unknown next hop} \xrightarrow{PoR, C_{i+1} \subseteq C_i} (s_{i+1}, C_{i+1}).$$

What persists across the gap is valid continuity, not possession by a pre-existing holder: a causal relationship to the predecessor and a non-expansive authority context. Proof of Relationship witnesses this as a property of the execution rather than of a holder; composed transitively under monotone attenuation ($C_{i+1} \preceq C_i$), it is Proof-of-Continuity, the propagation invariant of the PIC model.

Heterogeneous operation spaces. The notation assumes a common privilege universe $O \times R$. In heterogeneous systems different hops use different vocabularies; let $O_i \times R_i$ be the universe at hop i . A transition is then governed by a policy translation

$$\mathcal{T}_{i \rightarrow i+1} : \mathcal{P}(O_i \times R_i) \rightarrow \mathcal{P}(O_{i+1} \times R_{i+1}),$$

and monotonicity becomes $C_{i+1} \subseteq \mathcal{T}_{i \rightarrow i+1}(C_i)$: the successor authority is no greater than what the predecessor permits under the local translation. We treat $\mathcal{T}$ as fixed policy input; synthesizing or verifying such translations is out of scope. Heterogeneity here is semantic rather than merely administrative: different hops may describe authority through different operation/resource vocabularies even within a single trust perimeter. In heterogeneous systems, safety is therefore relative to the soundness of the chosen policy translation $\mathcal{T}_{i \rightarrow i+1}$; an overly permissive translation is a policy error rather than a violation of the continuity invariant itself.

Lemma 1 (Continuity implies origin binding). *If $PoC(\pi)$ holds, then s_n is causally linked to s_0 through a verifiable relational chain and $C_n \subseteq C_0$.*

Proof. Each $PoR(s_i, s_{i+1})$ links consecutive steps; their composition links s_n to s_0 . Non-expansiveness at each hop gives $C_n \subseteq \dots \subseteq C_0$.

We state the scope explicitly: this paper models the continuity invariant; the concrete (e.g. cryptographic) construction of PoR is out of scope and is realized by a companion enforcement architecture. As capability and token models assume their credentials unforgeable, we assume the lineage is unforgeable.

5.1 Authorization Rule

The following rule is evaluated only for a valid continuity chain, i.e. a chain in which every adjacent transition satisfies both $PoR(s_i, s_{i+1})$ and $C_{i+1} \subseteq C_i$.

For the common privilege universe $O \times R$, (o, r) is authorized at the final hop iff:

$$(o, r) \in \bigcap_{i=0}^n C_i.$$

Since the chain is monotonic decreasing:

$$C_n \subseteq C_{n-1} \subseteq \dots \subseteq C_0,$$

we have:

$$\bigcap_{i=0}^n C_i = C_n.$$

Thus no hop may acquire authority not already present at the origin, and any authority dropped at some hop j is lost irreversibly: by monotonicity it cannot reappear at any later hop.

5.2 The Ontological Shift: Possession vs Continuity

Possession- and capability-based models [2, 3] answer a *spatial* question: does an actor hold authority over an object? Authority is a point in $O \times R$. This model is atemporal: it has no axis on which to record the execution that led to an action.

Proof-of-Relationship supplies the missing axis: it binds an execution step to its causal predecessor. Proof-of-Continuity is built on it — continuity is the transitive composition of relationships — and combines two conditions: monotone restriction of privileges ($C_{i+1} \subseteq C_i$) and causal continuity, via PoR , of the execution that carries them.

Authority therefore no longer lives in $O \times R$ alone but in $O \times R \times \mathcal{L}$, where $\mathcal{L}$ is the causal lineage of the execution. Concretely, $\mathcal{L}$ is the set of finite linear causal lineages: a lineage is a sequence $\ell = \langle s_0 \prec s_1 \prec \dots \prec s_n \rangle$, where $\prec$ denotes causal precedence, witnessed between adjacent steps by PoR , so each event is an occurrence of a privilege within a particular lineage. Two actions identical in possession — same principal, same token, same privilege — may be distinct in continuity, because they descend from different causes. The same (o, r) exercised as a valid continuation of an authorizing request, and exercised by an executor drawing on an unrelated authority it happens to hold, is indistinguishable under possession, yet the two are distinct executions:

one continues an authorizing lineage, the other continues none. Possession is the projection onto $O \times R$ that forgets the lineage axis. Possession is the spatial component of authority; lineage is its temporal component, and a possession-only model is therefore a projection of authority that forgets time. This does not make possession-based models wrong; it makes them incomplete for the multi-hop case, where authority continuation becomes a temporal property. Thus the relevant event is not merely (p, o, r) , or identity plus privilege, but an occurrence of a privilege within a lineage. When the same principal and the same privilege occur in two different lineages, the identity coordinate does not separate them; the lineage coordinate does.

In the projection language this is the *temporal reading* of the confused deputy of Definition 2: an action spatially valid but temporally invalid — the deputy holds (o, r) , yet $(o, r) \notin C_0$ for the request that caused it — a collision under the projection that forgets $\mathcal{L}$, two distinct executions that possession cannot tell apart. Section 6 gives the operational characterization (the *open passthrough*), which refines Definition 2 with designation.

Relation to capabilities. Capabilities solve the spatial question and more: by fusing designation and authority in the act of invocation, a capability binds authority to a single causal step — in our terms *single-hop continuity*, a Proof-of-Relationship — so the classical confused deputy is ruled out, not merely reduced. What a capability treated purely as a held artifact lacks is the extension of that binding across a chain: applied outside the lineage that requested it, it falls back to lineage-invariant possession.

Proof-of-Possession and Proof-of-Continuity are two authority-propagation primitives. A capability invocation is a one-hop instance of the continuity primitive, whereas a capability held and presented as an artifact rests on possession; a capability system that carried continuity along the whole chain would, in these terms, be PIC-compliant. PIC is thus a temporal refinement of capability-style propagation, not a replacement — a dimension possession alone cannot encode. Both eliminate the confused deputy by construction rather than by runtime vigilance, but over different spans. Within the object-capability model, fusing designation and authority at the point of invocation removes the classical, single-hop mismatch structurally: a deputy cannot apply authority to a target it was not given authority for. The mismatch re-appears only when the capability model is left — under ambient authority, or when a capability is used as a held artifact outside the occurrence that carried it, the capability-as-possession case analysed above, where authority reverts to possession. PIC extends the same by-construction

guarantee across a multi-hop lineage: the confused deputy is not a state a valid PIC execution can represent at any hop, since a privilege absent from the origin authority context is not a valid later state at all.

5.3 Projection: Lineage-Invariant Possession Cannot Individuate Occurrences

We make the projection precise. An *event* is an occurrence exercising a privilege within a lineage,

$$e = (o, r, \ell) \in E := O \times R \times \mathcal{L},$$

and $p : E \rightarrow O \times R$, $p(o, r, \ell) = (o, r)$, is the projection that forgets the lineage.

Definition 7 (Possession policy). *An authorization policy $A : E \rightarrow \{0, 1\}$ is possession-based iff it is invariant under lineage: it factors through p , i.e. there exists $\bar{A} : O \times R \rightarrow \{0, 1\}$ with $A = \bar{A} \circ p$. Equivalently, $A(o, r, \ell) = A(o, r, \ell')$ for all $\ell, \ell' \in \mathcal{L}$: the verdict does not depend on which lineage produced the event.*

Theorem 3 (Lineage-invariant policies cannot individuate occurrences). *Let $e = (o, r, \ell)$ and $e' = (o, r, \ell')$ be two events with the same privilege but distinct lineage, $p(e) = p(e')$ and $\ell \neq \ell'$. Then every possession-based policy satisfies $A(e) = A(e')$.*

Proof. $A = \bar{A} \circ p$ and $p(e) = p(e')$, hence $A(e) = \bar{A}(p(e)) = \bar{A}(p(e')) = A(e')$.

The confused deputy is an instance of a forbidden pair: events sharing (o, r) but differing in lineage, one of which must be authorized and the other denied. One lineage originates from a context holding (o, r) ; the other exercises (o, r) though it never continued a context that held it. By the theorem, *no* lineage-invariant possession-based policy can separate them.

Theorem 4 (Any resolution reintroduces continuity). *A policy A authorizes e and denies e' , with $p(e) = p(e')$ and $\ell \neq \ell'$, only if A is not lineage-invariant; that is, A must read some function $g : \mathcal{L} \rightarrow X$ of the lineage with $g(\ell) \neq g(\ell')$.*

Proof. If A were lineage-invariant it would factor through p and give $A(e) = A(e')$ by the previous theorem, contradicting $A(e) \neq A(e')$. Hence A depends on the lineage coordinate: some $g(\ell)$ distinguishes e from e' .

Such a g is exactly a continuity mechanism — a proof-of-relationship extract of the lineage. The mitigations deployed within possession-based systems (nonces, idempotency keys, timestamps, audience and context binding) are concrete instances of g : they resolve the confused deputy only by reintroducing lineage information — that is, continuity — into the decision. Whether the lineage is carried alongside the request or embedded inside the artifact is immaterial: a token that encodes a nonce or a causal context makes A depend on ℓ and is therefore continuity-aware by definition, not possession-based. The distinction is not where the information lives but whether the decision reads it. The limitation is not an engineering failure of possession-based systems; it is a modeling boundary of lineage-invariant authorization: by the first theorem it is structurally unable to individuate occurrences, and every working fix is lineage-sensitive — continuity-aware — rather than lineage-invariant.

Why the possession model is representative. In the abstraction considered here, a bearer token used without explicit context or lineage binding is lineage-invariant: the decision reads the artifact and the requested privilege, not the causal lineage of the use. A JWT, for instance, asserts that “the holder may read,” not the causal chain that led to it. Such systems often work not because they encode continuity explicitly, but because a perimeter supplies part of it implicitly — mutual TLS, network boundaries, and transport-inferred delegation often stand in for the missing lineage. Once the perimeter dissolves — zero-trust deployments, multi-organisation chains on the public internet, and autonomous AI agents calling tools and services — lineage-invariance becomes visible as a vulnerability at the second hop. The definition of a possession policy above therefore describes deployed systems, not a caricature of them. A trust perimeter may hide the temporal gap by making authority and causality appear co-located, but the model does not depend on crossing such a perimeter: the mismatch can arise whenever authority is selected independently of the execution that caused its use.

5.4 A Trade-off Theorem for Authority Propagation

The projection of Section 5.3 is the technical basis for a trade-off among three properties an authority-propagation system may be asked to satisfy at once.

Definition 8 (Lineage-invariant authorization). *Authorization of an event $e = (o, r, \ell)$ depends only on the possessed authority for (o, r) , not on the lineage ℓ that caused the event.*

Definition 9 (Authority-mixing-capable delegation). *An executor may hold both a request-derived authority context C_0 and an independent authority source C_D , with some (o, r) such that $(o, r) \notin C_0$ and $(o, r) \in C_D$.*

Definition 10 (Confused-deputy safety). *No request whose origin context C_0 lacks (o, r) can cause a later executor to exercise (o, r) using authority not derived from that request.*

Theorem 5 (Possession–delegation–safety trade-off). *No authority-propagation system can simultaneously satisfy lineage-invariant authorization, authority-mixing-capable delegation, and confused-deputy safety.*

Proof. Suppose all three hold. By authority-mixing-capable delegation there is an executor D with a request-derived context C_0 and an independent source C_D , and an (o, r) with $(o, r) \notin C_0$ and $(o, r) \in C_D$. Let a request cause D to process a target requiring (o, r) . Lineage-invariant authorization reads that D possesses C_D granting (o, r) ; but by the projection theorem of Section 5.3, events sharing (o, r) yet differing in lineage receive the same judgment, so the policy cannot condition on whether C_D belongs to the lineage of this request. The system therefore admits an execution in which the request causes D to exercise (o, r) while $(o, r) \notin C_0$, violating confused-deputy safety — a contradiction.

The theorem is not an attack on possession: it shows that lineage-invariant possession is insufficient for multi-hop authority propagation precisely when authority mixing and confused-deputy safety are both required. Since the three cannot hold together, a system must relinquish exactly one, and only one such choice is acceptable. Forbidding authority-mixing-capable delegation would bar executors from holding authority of their own alongside request-derived authority — impractical in real deployments, where a service legitimately retains its own privileges. Accepting confused-deputy executions abandons the safety property at issue. The remaining option is to relinquish lineage-invariance — to make authorization lineage-sensitive. Proof-of-Continuity takes this third option: it does not remove possession but restores the missing temporal coordinate, so that authorization can read whether each authority use is a valid continuation of the context that caused it. The contribution is not to reject possession but to show that possession is the spatial projection of a richer authority relation; the missing coordinate is temporal lineage.

Corollary 1 (Continuity restores confused-deputy safety). *Relinquishing lineage-invariance regains safety: under non-expansive propagation $C_{i+1} \subseteq$*

Property relin- quished	Consequence	Outcome
Authority-mixing delegation	executors may not hold authority independent of the request	rejected (impractical)
Confused-deputy safety	confused-deputy executions be- come admissible	rejected (unsafe)
Lineage-invariance	authorization becomes lineage- sensitive	adopted (Proof-of-Continuity)

Table 1: The three properties cannot hold together, so a system must relinquish one. Forbidding authority mixing is impractical and accepting confused deputies is unsafe; the only acceptable choice is to relinquish lineage-invariance — the option adopted by Proof-of-Continuity.

C_i , so $C_k \subseteq C_0$, and an $(o, r) \notin C_0$ cannot appear at any later hop. This is the impossibility established in Section 7.

6 Safety Property

Definition 11 (Origin-bounded authority). *A model enforces origin-bounded authority if every executable privilege at hop n was originally granted at hop 0.*

We refine Definition 2 to separate legitimate delegation from the confused deputy. The distinguishing variable is not the authority source but whether the action’s target is designated by the request and whether the requester was entitled to bring it about.

Definition 12 (Authority to cause). *The origin context C_0 records the authority the requester is entitled to cause, not only to perform directly. A later action belongs to the requester’s lineage iff it realizes an authority in C_0 ; otherwise it is the deputy’s own action, rooted in a distinct origin (the deputy itself) and unconstrained by C_0 .*

Definition 13 (Open passthrough). *A deputy is an open passthrough when, as a consequence of a request, it applies authority to a target designated by that request for an action the requester was not entitled to cause: $(o, r) \notin C_0$ yet (o, r) executes because of the request. This is exactly the confused deputy.*

These separate three cases. (i) A deputy’s own housekeeping — the compiler writing its statistics file — is rooted in the deputy as origin and

is unconstrained by C_0 . (ii) Legitimate delegation — a payment gateway moving funds because the user was entitled to cause the payment — realizes an authority in C_0 and is a valid continuation, even though the user does not directly hold the banking authority. (iii) The confused deputy — writing a user-designated file the user could not cause — is an open passthrough, the one case the model forbids. The relevant question is not whether the origin could perform the downstream operation directly, but whether the origin was authorized to *cause* it within this lineage. When successive services use different operation spaces, this is mediated by the policy translation $\mathcal{T}$ introduced above, whose synthesis and verification are left to future work.

Example. If $C_0 = \{(\text{convert}, r)\}$, then (write, r) can never be authorized within the same execution chain: $(\text{write}, r) \notin C_0$ and $C_{i+1} \subseteq C_i$ at every transition, so $(\text{write}, r) \notin C_n$.

7 Impossibility of the Confused Deputy

The confused deputy condition requires an authority mismatch: a privilege absent from the authority context of the request is nevertheless exercised as a consequence of that request.

This is the paper’s second impossibility result. Section 5.3 shows that possession *cannot* individuate occurrences; here we show the dual: under continuity the confused deputy is *not a state the model can represent*. The condition is unsatisfiable within any valid PIC execution — an impossibility by construction, at the level of valid model behavior.

Theorem 6 (Confused deputy is impossible under PIC). *In any valid PIC execution — one whose adjacent transitions satisfy PoR and monotonicity $C_{i+1} \subseteq C_i$, so that authority is origin-bounded ($C_n \subseteq C_0$) — the confused deputy conditions cannot be jointly satisfied as valid behavior within the model.*

Proof. By PoR the hops form a single lineage rooted at s_0 , and by monotonicity $C_k \subseteq C_0$ for all k ; hence $(o, r) \notin C_0$ implies $(o, r) \notin C_k$. Authority the deputy holds outside this lineage has a distinct origin (Section 6) and is not exercised as part of the request; within the request’s lineage the privilege mismatch a confused deputy requires is therefore unsatisfiable.

Interpretation. Authority in PIC is not something that a subject merely *has*; it is a continuity property of the execution chain. Since no hop may

introduce or reinterpret authority absent at the origin, a confused-deputy execution may still be attempted physically, but it cannot be accepted as valid behavior within the PIC model.

8 Discussion

PoP systems intentionally derive authorization from artifact control, which is often useful and necessary for local authorization decisions. In multi-hop execution, however, artifact control alone does not determine whether a later use is a valid continuation of the authority lineage that caused it. PIC/PoC systems instead propagate only non-expansive subsets of authority along a causally verified execution chain.

Proof of Possession proves artifact control. Proof of Relationship proves execution relationship. Proof of Continuity proves authority continuation. Proof-of-Possession establishes control over an authority artifact at a point in execution; it does not by itself establish that a later use is a valid continuation of the authority lineage that caused it, which is the temporal condition Proof-of-Continuity adds.

Authority is continuity. Under PIC, authority is not a static property an entity merely *has* but a continuity property of an execution: a privilege is authorized at a step iff it is a non-expansive continuation of the context that caused it. In this sense authority *is* continuous — it exists only as the ongoing continuation of an origin, not as a possession detached from its lineage.

Design space. The temporal binding may be realized in two ways: carried *inside* the authority artifact itself, as attenuated and context-bound capabilities, or maintained *alongside* it as an explicit continuity proof. Either is continuity-aware: by Theorem 4 what matters is not where the binding lives but whether the authorization decision reads it. An artifact whose binding is ignored — presented outside the occurrence that carried it — reverts instead to lineage-invariant possession. PIC takes the latter position, treating continuity as a first-class property of the execution chain; the two approaches are complementary rather than exclusive.

The resulting security invariant is simple: no downstream service can perform, as part of a valid execution chain, an operation that was not authorized at the origin. This is a modeling refinement rather than a criticism of existing mechanisms: possession, capabilities, bearer tokens, and

proof-of-possession bindings remain useful local authorization tools, while PoC makes explicit the additional temporal condition needed when authority is propagated across an execution lineage.

Authority as a distributed transaction. A multi-hop authority chain is a distributed transaction over authority rather than over data. A possessed artifact enables a local authorization decision at the point of use — often necessary for deployability, performance, and loose coupling — but revocation, delegation state, scope changes, and lineage are global authority facts. Local authorization and global authority consistency diverge when these facts are not represented at the point of use. PoC addresses this by carrying lineage explicitly and requiring non-expansive authority propagation across the chain.

Continuity is therefore not identity propagation. Identity providers, wallets, OIDC/OIDC4VC-style flows [13, 14], and WIMSE/SPIFFE-style workload identity [11, 12] remain important for authentication, credential presentation, audit, accountability, and origin bootstrap; but an identifier repeated at later hops does not by itself prove that the authority being exercised belongs to the same execution lineage. A lineage begins when a permissioned entity expresses an intent within its privileges, creating an origin authority context C_0 ; subsequent hops need not carry or reinterpret the origin identity, but must preserve a non-expansive authority context causally linked to that origin.

Application to AI agents. For authority propagation, AI agents are a special case of autonomous distributed execution; we name them because the temporal gap is most acute there. An agent runs a chain of steps — planning, tool invocations, downstream calls — while holding multiple authority sources at once: its own service credentials, delegated user tokens, and per-tool scopes. The classical local-filename designation reappears as an agent-influenced parameter — a tenant id, document id, role ARN, queue message, or request target. When the caller influences this target and the agent satisfies it using authority from a different context, the result is the distributed confused deputy. PIC binds each tool call and downstream request to the causal lineage that triggered it, so an agent cannot apply authority outside the execution that caused the action. An agent asked to summarize a document carrying an indirect prompt injection — instructions to delete files or exfiltrate secrets — may attempt those actions, but under PoC they are valid only if the origin authority context holds the corresponding privileges:

if it authorizes only $(\text{summarize}, d)$, then (delete, r) or $(\text{exfiltrate}, r)$ cannot be valid continuations.

Related Work

The confused deputy problem was introduced by Hardy [1]. Capability-based access control [2] and the analysis of the confused deputy under capabilities [3] address the spatial dimension of authority — whether an actor holds authority over an object. This paper isolates the orthogonal temporal dimension: whether the authority used belongs to the execution that caused the action. We take the confused deputy not as the specific operating-system compiler scenario in which it was first described, but as the general authority/causality mismatch that scenario exemplifies, and recast it for discrete multi-step execution — the setting of distributed systems, service chains, and AI agents rather than a single local invocation. Framed this way, the confused deputy is one instance of a class of authority-propagation vulnerabilities that share the same mismatch; under Proof-of-Continuity that mismatch cannot arise as valid model behavior, since a privilege absent from the origin authority context cannot be exercised later in the same lineage. The paper compares possession-based authority propagation with provenance-continuity-based authority propagation on exactly this axis.

The closest prior work is history-based access control [4], where the authority available to a computation depends on the history of what has executed. We share the premise that authority is a function of execution rather than of static possession. Unlike a retrospective check over past callers (as in stack inspection), PoC is a forward propagation discipline: each hop must be causally related to its predecessor and must not expand the authority it received, including across asynchronous and cross-boundary steps. The contribution here is orthogonal and more specific: we frame the dependence as a *dimension* $\mathcal{L}$ orthogonal to the privilege (o, r) , prove that possession (any policy invariant under $\mathcal{L}$) cannot individuate occurrences, and show that the confused deputy is a collision under the projection that forgets $\mathcal{L}$ — so that possession-based mitigations that distinguish such cases must become lineage-sensitive, hence continuity-aware. The object-capability literature [5] analyses the confused deputy as a separation of designation from authority, which capabilities reunify; this is the spatial resolution. Our result adds the temporal axis: in the abstraction of this paper, a capability treated only as a portable held artifact, without occurrence or lineage binding, falls into the lineage-invariant projection and cannot distinguish two occurrences that

differ only in lineage.

Bearer tokens, macaroons, and proof-of-possession bindings.

OAuth 2.0 includes bearer-token usage as a common authorization pattern [6, 7]. A bearer token used without explicit context or lineage binding supports a decision that reads the token and (o, r) but not the causal lineage of the use: it is lineage-invariant, and thus subject to the first theorem. Macaroons [8] add *caveats* that only attenuate and chain cryptographically — precisely our monotone attenuation $C_{i+1} \preceq C_i$ — yet a macaroon remains a bearer credential: absent a caveat binding the request context, it can be presented outside the lineage that produced it. DPoP [9] and Token Binding [10] bind a token to a key or transport channel; in the terms of Theorem 4 such bindings make the decision read additional occurrence context, and are continuity-aware in a local sense. Each reduces cross-context misuse to the extent that a caveat or binding makes the decision read ℓ ; none by itself establishes global Proof-of-Continuity, since the decision still need not read the full causal lineage from the origin across a multi-hop chain. PIC makes that lineage dimension global and explicit.

The PIC model is developed independently in [17], where Provenance Identity Continuity is introduced as a general model for distributed execution systems. The present paper extends an earlier version [18], refining it with the relationship/continuity distinction and the spatial/temporal analysis of authority.

9 Limitations and Future Work

This paper presents a minimal linear model of Proof-of-Continuity. Several of the directions below are not defects of the model but properties to be developed within the same continuity principle; we distinguish genuine scope limits from natural extensions. First, revocation is not treated as a retroactive operation on an already issued lineage: a practical system must decide whether revocation blocks only future transitions, invalidates outstanding continuations, or triggers lineage-wide cancellation. Second, composition: the model is linear, and forks, joins, fan-out, fan-in, and DAG-shaped executions are a natural generalization of $\mathcal{L}$ to a partial order — an extension of the continuity principle to composed lineages rather than a limitation of it. Third, the construction of *PoR* is abstract: concrete implementations may use signatures, hash chains, verifiable credentials [16], secure channels, trusted execution environments, or tamper-evident logs,

with implementation-dependent security and performance. Fourth, lineage validation is an implementation concern rather than a limitation of the model: practical realizations need not revalidate an entire history at every hop, and may use checkpoints, summarized proofs, compact encodings, or accumulator-style constructions to keep validation cost bounded. Finally, heterogeneous systems require policy translations $\mathcal{T}$ between local vocabularies, whose synthesis, verification, and governance we leave to future work.

10 Conclusion

This paper introduced Proof-of-Continuity as a minimal temporal model for authority propagation. The core distinction is between possession, which proves control over an authority artifact; relationship, which proves causal linkage between adjacent execution steps; and continuity, which proves that authority has propagated without expansion across the whole lineage. The main result is that lineage-invariant possession cannot distinguish occurrences that share the same privilege but arise from different causes; conversely, under Proof-of-Continuity a privilege absent from the origin authority context cannot appear later in the same valid execution chain. The confused deputy condition is therefore not accepted as valid model behavior — it is precisely the authority/causality mismatch that continuity rules out. More broadly, PoC reframes authority propagation as a temporal property of an execution lineage rather than merely a static property of artifact possession.

Note on the name PIC. In Provenance Identity Continuity, the term Identity refers to the security identity or execution identity of the permissioned entity from which authority is created or anchored at the origin. It does not mean that identity is propagated as the authorization primitive at every hop. The model instead shifts the authorization burden from repeatedly interpreting identity to verifying execution continuity: a causal lineage carrying a non-expansive authority context. Provenance denotes the causal origin and lineage of the execution, Identity denotes the authenticated permissioned origin, and Continuity denotes the preservation of authority across later hops without expansion. Identity and identifiers therefore remain essential for authentication, credential presentation, audit, accountability, and origin bootstrap; PoC concerns the authorization continuity that must hold after that origin has been established.

Acknowledgments

This paper follows an earlier version of the PIC Model made available on Zenodo in December 2025 [17]. The author thanks Alan H. Karp, whose experience was valuable in stress-testing the concepts of Proof-of-Continuity and Proof of Relationship. The concepts presented here are the author’s own, and this acknowledgment does not imply his endorsement of the paper. The author also used automated language assistance tools for editorial refinement, LaTeX formatting, and notation polishing. All conceptual contributions, models, and proofs are the author’s own.

References

- [1] N. Hardy. The Confused Deputy: Or Why Capabilities Might Have Been Invented. *ACM SIGOPS Operating Systems Review*, 22(4):36–38, 1988.
- [2] J. B. Dennis and E. C. Van Horn. Programming Semantics for Multiprogrammed Computations. *Communications of the ACM*, 9(3):143–155, 1966.
- [3] M. S. Miller, K.-P. Yee, and J. S. Shapiro. Capability Myths Demolished. Technical Report SRL2003-02, Systems Research Laboratory, Johns Hopkins University, 2003.
- [4] M. Abadi and C. Fournet. Access Control Based on Execution History. In *Proceedings of the Network and Distributed System Security Symposium (NDSS)*, 2003.
- [5] M. S. Miller. Robust Composition: Towards a Unified Approach to Access Control and Concurrency Control. PhD thesis, Johns Hopkins University, Baltimore, MD, USA, 2006.
- [6] D. Hardt, editor. The OAuth 2.0 Authorization Framework. RFC 6749, Internet Engineering Task Force, October 2012.
- [7] M. Jones and D. Hardt. The OAuth 2.0 Authorization Framework: Bearer Token Usage. RFC 6750, Internet Engineering Task Force, October 2012.
- [8] A. Birgisson, J. G. Politz, Ú. Erlingsson, A. Taly, M. Vrabie, and M. Lenczner. Macaroons: Cookies with Contextual Caveats for Decentral-

- ized Authorization in the Cloud. In *Proceedings of the Network and Distributed System Security Symposium (NDSS)*, 2014.
- [9] D. Fett, B. Campbell, J. Bradley, T. Lodderstedt, M. B. Jones, and D. Waite. OAuth 2.0 Demonstrating Proof of Possession (DPoP). RFC 9449, Internet Engineering Task Force, September 2023.
 - [10] A. Popov, M. Nyström, D. Balfanz, and J. Hodges. The Token Binding Protocol Version 1.0. RFC 8471, Internet Engineering Task Force, October 2018.
 - [11] IETF WIMSE Working Group. Workload Identity in Multi-System Environments (WIMSE). Internet Engineering Task Force, 2024. <https://datatracker.ietf.org/wg/wimse/>.
 - [12] SPIFFE Project. Secure Production Identity Framework for Everyone (SPIFFE) Specification. Cloud Native Computing Foundation, 2017. <https://github.com/spiffe/spiffe>.
 - [13] OpenID Foundation. OpenID Connect Core 1.0. 2014. https://openid.net/specs/openid-connect-core-1_0.html.
 - [14] OpenID Foundation. OpenID for Verifiable Credentials (OpenID4VC). Specification, 2024. <https://openid.net/sg/openid4vc/>.
 - [15] World Wide Web Consortium (W3C). Decentralized Identifiers (DIDs) v1.0. W3C Recommendation, 2022. <https://www.w3.org/TR/did-core/>.
 - [16] World Wide Web Consortium (W3C). Verifiable Credentials Data Model v1.1. W3C Recommendation, 2022. <https://www.w3.org/TR/vc-data-model/>.
 - [17] N. Gallo. PIC Model — Provenance Identity Continuity for Distributed Execution Systems. Zenodo, 2025. <https://zenodo.org/records/17777421>.
 - [18] N. Gallo. Authority Propagation Models: PoP vs PoC and the Confused Deputy Problem. Zenodo, 2025. <https://doi.org/10.5281/zenodo.17833000>.